UNCOVERING THE CLOUD: IMPORTANT DIFFICULTIES AND NEW CONCERNS IN SECURE COMPUTING

Group 2

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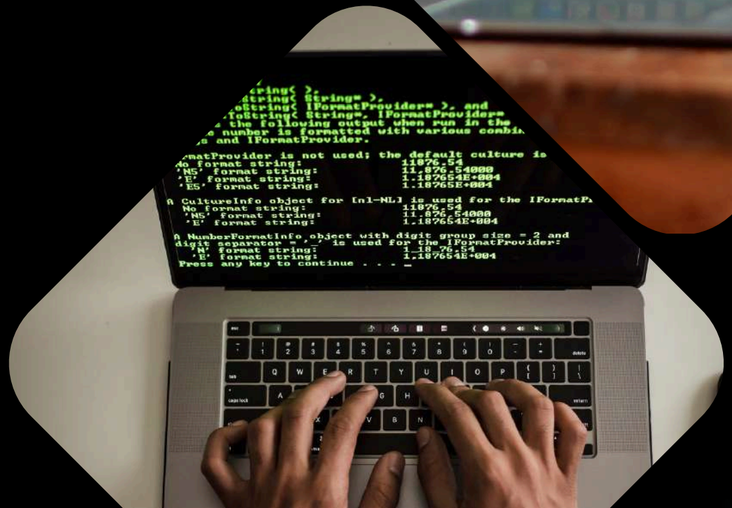
OUTLINE

- Introduction & Background
- Cloud Computing Fundamentals
- Cloud Security Threats
- Prevention Strategies
- Conclusion & Recommendations



INTRODUCTION & BACKGROUND

Why Cloud Security Matters

- Cloud computing has transformed IT infrastructure across industries
- Major providers: Amazon, Google, Microsoft expanding rapidly
- Offers flexibility, scalability, and cost-efficiency

Current Challenge

- Integration of third-party services creates new vulnerabilities
- Data security and privacy becoming increasingly complex
- Need for comprehensive security strategies

CLOUD COMPUTING FUNDAMENTALS

What is Cloud Security?

Definition: Set of policies, technologies, and controls designed to protect:

- Data
- Applications
- Infrastructure hosted in cloud environments

Core Goals:

- Confidentiality (Data protection)
- Integrity (Data accuracy)
- Availability (Service accessibility)

Unique Cloud Challenges

- Multi-tenancy environments
- Virtualization complexities
- Third-party service dependencies
- Shared responsibility models

CLOUD DEPLOYMENT MODELS

Public Cloud

Third-party providers deliver resources over internet

Advantages:

- Cost effective (pay-per-use)
- Highly scalable
- Quick deployment
- No hardware maintenance

Disadvantages:

- Less control over environment
- Potential compliance issues
- Vendor lock-in risks

Examples: AWS, Microsoft Azure, Google Cloud

Private Cloud

Dedicated infrastructure for single organization

Advantages:

- Full control and customization
- Enhanced security and isolation
- Performance stability
- Better compliance support

Disadvantages:

- Higher costs
- Slower scalability
- More management responsibility

MAJOR CLOUD SECURITY THREATS

Technical Vulnerabilities

- Insecure APIs
  - Weakly secured application interfaces
  - Entry points for unauthorized access
- Shared Technology Vulnerabilities
  - Multi-tenant environment risks
  - Cross-tenant contamination potential
- Unknown Risk Profiles
  - Third-party service dependencies
  - Limited visibility into security practices

Human-Driven Risks

- Data Abuse and Leakage
  - Misuse of stored information
  - Poor access control implementations
- Malicious Insiders
  - Employees/contractors with privileged access
  - Authority misuse potential
- Account/Service Hijacking
  - Stolen credentials via phishing
  - Brute force attacks
  - Unauthorized service manipulation

Service Disruption

- Denial of Service (DoS/DDoS) Attacks
  - Service overloading
  - Resource exhaustion
  - Business continuity threats

PREVENTION STRATEGIE

Technical Defenses

- Encryption: Protect data at rest and in transit
- Multi-Factor Authentication (MFA): Additional security barriers
- Regular Security Audits: Proactive vulnerability identification
- Secure Configuration Management: Prevent misconfigurations

Organizational Measures

- Access Control & Identity Management
  - Role-based access controls
  - Principle of least privilege
- Incident Response Planning
  - Fast breach containment
  - Effective recovery procedures

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Emerging Technologies

- **AI-based Monitoring:** Advanced threat detection
- **Blockchain Integration:** Decentralized security models
- **Threshold Cryptography:** Enhanced key management

RECOMMENDATIONS

For Organization

- Implement comprehensive security frameworks
- Regular security assessments and audits
- Invest in employee training programs
- Develop incident response capabilities
- Choose reputable cloud providers with transparency

For Users

- Verify email senders and links
- Protect private credentials
- Only share information with verified institutions
- Stay updated on security threats

FUTURE CONSIDERATIONS

Emerging Trends

- Zero Trust Architecture: Verify everything, trust nothing
- Cloud-Native Security Tools: Built for cloud environments
- Automated Threat Response: AI-driven security operations
- Quantum-Resistant Encryption: Future-proofing data protection

Research Directions

- Advanced cryptographic techniques
- Behavioral analysis for insider threats
- Cross-cloud security standards
- Privacy-preserving technologies

CONCLUSION

- Traditional security models don't work for cloud environments due to new risks like shared systems and insider threats
- Success requires comprehensive defense combining technical controls and organizational measures
- Future security depends on three pillars: advanced technology, clear governance, and user training
- Organizations integrating all elements maximize cloud benefits while maintaining security

THANK YOU